

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain construction, working and advantages of Ultrasonic Flow meter.
- Q.37 Describe construction and working of Ionization Gauge.
- Q.38 Explain level measurement using capacitive transducer with diagram.

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5th Sem / IC, EI
Sub : Process Instrumentation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Thermocouple works on _____ effect.
a) Hall b) Seebeck
c) Doppler d) Piezo
- Q.2 Pirani gauge is used for measurement of _____.
a) High Pressure
b) Low Pressure
c) Medium Pressure
d) Atmospheric Pressure
- Q.3 Rotameter is used to measure _____.
a) Temperature b) Pressure
c) Flow d) Level
- Q.4 Pascal is the unit of _____.
a) Temperature b) Pressure
c) Flow d) Density
- Q.5 Ultrasonic flow meter works on _____ principle.
a) Seebeck b) Doppler Effect
c) Hall Effect d) Radiation

- Q.6 Bridgman gauge is used for ____ pressure
 a) Low b) High
 c) Medium d) Vacuum
- Q.7 RTD is used for measurement of ____
 a) Flow b) Level
 c) Temperature d) Pressure
- Q.8 Orifice meter is a ____ flow meter
 a) Positive displacement
 b) Differential pressure
 c) Turbine
 d) Magnetic
- Q.9 Unit of frequency is ____
 a) Ohm b) Pascal
 c) Hz d) Henry
- Q.10 Capacitive transducer is used for ____ measurement
 a) Level b) Temperature
 c) Flow d) Density

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Pressure.
 Q.12 Write any two units of temperature.
 Q.13 Thermistors are highly non-linear. (T/F)
 Q.14 define Seebeck effect.
 Q.15 Ionization gauge measures ____ pressure.

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- Q.16 Write relation between absolute and gauge pressure.
 Q.17 Manometer is used for pressure measurement. (T/F)
 Q.18 Write any one advantage of thermistor.
 Q.19 Define Flow.
 Q.20 Gamma rays are used for level measurement. (T/F)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain working principle of Thermocouple.
 Q.22 Discuss Orifice flow meter with advantages and disadvantages.
 Q.23 Explain Pirani gauge for low pressure measurement.
 Q.24 Describe Electromagnetic flow meter.
 Q.25 Explain capacitive methods for level measurement.
 Q.26 Discuss Radiation Pyrometer with advantages.
 Q.27 Describe construction and working of Rotameter.
 Q.28 Explain Bridgman gauge.
 Q.29 Discuss Ultrasonic method for level measurement.
 Q.30 Write short note on Fiber Optic Thermometer.
 Q.31 Explain Bellows pressure measurement.
 Q.32 Describe Hot wire anemometer.
 Q.33 Explain U-tube manometer.
 Q.34 Discuss Turbine flow meter.
 Q.35 Write short note on IR detector.

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